Protocol/Statistical Analysis Plan: I8B-MC-ITRT

Effect of Injection Site on the Relative and Absolute Bioavailability of Single Dose of LY900014 in Healthy Subjects

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Approval Date:

Protocol I8B-MC-ITRT(a)
Effect of Injection Site on the Relative and Absolute Bioavailability of Single Dose of LY900014 in Healthy Subjects

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LY900014

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Table of Contents

Effect of Injection Site on the Relative and Absolute Bioavailability of Single Dose of LY900014 in Healthy Subjects

Section **Page**

Protocol I8B-MC-ITRT(a) Effect of Injection Site on the Relative and Absolute Bioavailability of Single Dose of LY900014 in Healthy Subjects 1
Table of Contents 2
1. Protocol Synopsis 8
2. Schedule of Activities 10
3. Introduction 14
3.1. Study Rationale 14
3.2. Background 14
3.3. Benefit–Risk Assessment 15
4. Objectives and Endpoints 17
5. Study Design 18
5.1. Overall Design 18
5.2. Number of Participants 19
5.3. End of Study Definition 19
5.4. Scientific Rationale for Study Design 20
5.5. Justification for Dose 20
6. Study Population 22
6.1. Inclusion Criteria 22
6.2. Exclusion Criteria 23
6.3. Lifestyle and/or Dietary Requirements 24
6.3.1. Meals and Dietary Restrictions 25
6.3.2. Caffeine, Alcohol, and Tobacco 25
6.3.3. Activity 25
6.4. Screen Failures 25
7. Treatment 26
7.1. Treatment Administered 26
7.1.1. Packaging and Labeling 26
7.2. Method of Treatment Assignment 26
7.2.1. Selection and Timing of Doses 27
7.3. Blinding 27

7.4. Dose Modification 27
7.5. Preparation/Handling/Storage/Accountability 27
7.6. Treatment Compliance 27
7.7. Concomitant Therapy 28
7.8. Treatment after the End of the Study 28
8. Discontinuation Criteria 29
8.1. Discontinuation from Study Treatment 29
8.1.1. Discontinuation of Inadvertently Enrolled Subjects 29
8.2. Discontinuation from the Study 29
8.3. Subjects Lost to Follow-up 29
9. Study Assessments and Procedures 30
9.1. Efficacy Assessments 30
9.2. Adverse Events 30
9.2.1. Serious Adverse Events 31
9.2.1.1. Suspected Unexpected Serious Adverse Reactions 31
9.2.2. Complaint Handling 32
9.3. Treatment of Overdose 32
9.4. Safety 32
9.4.1. Laboratory Tests 32
9.4.2. Vital Signs 32
9.4.3. Electrocardiograms 32
9.4.4. Other Tests 33
9.4.4.1. Physical Examinations 33
9.4.4.2. Body Weight 33
9.4.4.3. Hip and Waist Circumference 33
9.4.5. Safety Monitoring 33
9.4.5.1. Glucose Monitoring 34
9.4.5.1.1. Severe Hypoglycemia 34
9.4.6. Injection-Site Assessments (Local Tolerability) 35
9.5. Pharmacokinetics 35
9.5.1. Bioanalysis 35
9.6. Glucodynamics (Euglycemic Glucose Clamp) 35
9.6.1. Immunogenicity Assessments 36
9.7. Genetics 37
9.8. Exploratory Biomarkers 37
9.9. Health Economics 37
10. Statistical Considerations and Data Analysis 38
10.1. Sample Size Determination 38

10.2. Populations for Analyses 38
10.2.1. Study Participant Disposition 38
10.2.2. Study Participant Characteristics 38
10.3. Statistical Analyses 38
10.3.1. Safety Analyses 38
10.3.1.1. Clinical Evaluation of Safety 38
10.3.1.2. Statistical Evaluation of Safety 39
10.3.2. Pharmacokinetic Analyses 39
10.3.2.1. Pharmacokinetic Parameter Estimation 39
10.3.2.2. Pharmacokinetic Statistical Inference 40
10.3.3. Glucodynamic Analyses 40
10.3.3.1. Glucodynamic Parameter Estimation 40
10.3.3.2. Glucodynamic Statistical Inference 40
10.3.4. Pharmacokinetic/Pharmacodynamic Analyses 40
10.3.5. Evaluation of Immunogenicity 41
10.3.6. Data Review during the Study 41
10.3.7. Interim Analyses 41
11. References 42

List of Tables

Table **Page**

Table ITRT.1. Objectives and Endpoints 17
Table ITRT.2. LY900014 Treatment Sequences 19
Table ITRT.3. Treatments Administered 26

List of Figures

Figure **Page**

Figure ITRT.1. Illustration of study design of Protocol I8B-MC-ITRT.....19

List of Appendices

Appendix **Page**

Appendix 1. Abbreviations and Definitions 43
Appendix 2. Clinical Laboratory Tests 47
Appendix 3. Study Governance, Regulatory, and Ethical Considerations 48
Appendix 4. Blood Sampling Summary 51
Appendix 5. Protocol Amendment I8B-MC-ITRT(a) Summary Effect of Injection Site on the Relative and Absolute Bioavailability of Single Dose of LY900014 in Healthy Subjects 52

1. Protocol Synopsis

Title of Study:
Effect of injection site on the relative and absolute bioavailability of single dose of LY900014 in healthy subjects.

Rationale:
In clinical practice, it is recommended that patients rotate insulin injections and may inject insulin at different tissue sites. The actual site of subcutaneous (SC) injection may influence the absorption pharmacokinetics (PK), as well as, the glucodynamic (GD) effect. The primary objective of this study is therefore to compare the bioavailability of insulin lispro following administration of the commercial formulation of LY900014 at 3 different injection sites SC in the abdominal wall, thigh, and deltoid. In addition, the intravenous (IV) administration will serve as the reference to determine the absolute bioavailability of SC injections into the deltoid, thigh or abdomen and GD activity.

Objectives/Endpoints:

Objectives	Endpoints
Primary	
To determine the relative bioavailability of insulin lispro from LY900014 after SC injection into the thigh and deltoid compared to the abdomen in healthy subjects.	Area under the concentration versus time curve from time zero to infinity (AUC[0-∞])
To determine absolute bioavailability of insulin lispro following SC administrations of LY900014 into the thigh, deltoid and abdomen compared to IV administration of LY900014 in healthy subjects.	Percent of the ratio of AUC(0-∞)
Secondary	
To evaluate the tolerability of LY900014 in healthy subjects.	Adverse events (AEs) and injection-site reactions
To evaluate the GD during euglycemic clamp procedure of LY900014 after SC injection into the abdomen, thigh and deltoid and IV in healthy subjects.	Total amount of glucose infused (G_{tot}) and maximum glucose infusion rate (R_{max}).

Summary of Study Design:

Study I8B-MC-ITRT is a single-center, open-label, 4-period, randomized, crossover, up to 10-hour euglycemic clamp study in healthy subjects to compare the insulin lispro PK and GD of LY900014 after SC administration of a 15-U injection at 3 different injection sites and a single IV bolus injection of 15 U. The SC injection sites are the abdomen, thigh, and deltoid. An IV injection into the forearm will be used to assess absolute bioavailability.

Treatment Arms and Planned Duration for an Individual Subject:
Subjects will be randomly assigned to 1 of 4 treatment sequences; each sequence will have 4 periods. In 3 of the 4 periods, subjects will be administered single SC doses of 15 U of LY900014 at 3 different injection sites, which include the thigh, abdomen and deltoid. One study period will include a single IV dose of 15 U of LY900014.

The study will include a 28-day screening period, followed by 4 study periods. There will be a wash-out period of at least 3 days between each study period. The follow-up visit will take place at least 14 days after the last dose.

Number of Subjects:
Up to 28 healthy men and women may be enrolled to target at least 22 subjects complete the study.

Statistical Analysis:

The primary statistical analyses for PK will be conducted on those subjects who receive at least 1 dose of study drug and have measurable insulin lispro concentrations. The primary statistical analysis for GD will be conducted on those subjects who complete at least 1 clamp procedure. Supportive analyses will be done on the key parameters for the subjects who complete all treatment periods with evaluable data.

Safety analyses will be conducted for all enrolled subjects, whether or not they complete all protocol requirements. All investigational product and protocol-procedure AEs will be listed, and if the frequency of events allows, safety data will be summarized using descriptive methodology. Safety parameters that will be assessed include safety laboratory parameters and vital signs, as well as evaluation of injection-site reactions. The parameters will be listed and may be summarized using standard descriptive statistics.

Log-transformed AUC(0-∞) and maximum observed serum concentration (C_{max}) for insulin lispro will be evaluated to estimate geometric least-squares means (LSmeans), ratios of geometric LSmeans between injection sites (deltoid and thigh compared with the abdomen), and their corresponding 90% confidence interval (CI) using the statistical model that includes injection site, period, and sequence as fixed effects, and subject within sequence as a random effect. The primary parameter for the statistical analysis of the relative bioavailability between injection sites will be the AUC(0-∞).

Log-transformed R_{max} and G_{tot} will be evaluated to estimate geometric LSmeans, ratios of geometric LSmeans between injection sites (deltoid and thigh compared with the abdomen), and their corresponding 90% CIs using the statistical model that includes injection site, period, and sequence as fixed effects, and subject within sequence as a random effect. For GD parameters that have at least 1 subject with a value equal to zero, a value equal to the smallest non-zero observed GD value for that parameter divided by 2 will be added to all values, and the analysis of the log-transformed data will be performed. In addition, as a sensitivity analysis, a nonparametric analysis will be performed for that parameter.

The PK and GD analyses will also be performed using the subset of the subjects who complete all treatment periods with evaluable data.

The frequency of antibody formation to insulin lispro will be determined. The relationship between the presence (or absence) of antibodies and AEs will be assessed. Likewise, the relationship between the presence of antibodies and the PK parameters and GD response to insulin lispro may be assessed.

2. Schedule of Activities

Study Schedule of Protocol I8B-MC-ITRT

Procedures	Screening	Periods 1, 2, 3, 4			FU/ ED[a]	Comments
	Up to Day -28	Day -1	Day 1	Day 2		
Informed consent	X					
Admission to CRU		X				
Enrollment		X				Enrollment prior to taking first predose blood sample for the study.
Overnight fast	X	X				Subjects are expected to fast for approximately 8 hours before screening, and at least 8 hours before each dose until the end of the glucose clamp procedure.
Height	X					
Weight	X	X[#]			X	[#] Period 1 only
Hip and waist circumference		X[#]				[#] Period 1 only
Medical history and physical examination	X					
Medical assessment	X		Predose	X*	X	Medical assessment includes medical review and physical examination, as appropriate. *Prior to discharge from CRU for each period
Vital signs (supine): blood pressure and pulse rate	X		Predose, 1 hour postdose (IV period only), and 10 hours postdose (at the end of clamp procedure)		X	Predose vital signs may be collected up to 3 hours before dosing.
12-Lead ECG	X		Predose and 10 hours postdose (at the end of clamp procedure)		X	Single ECGs will be collected for safety. Predose ECGs may be collected up to 3 hours before dosing.
Clinical laboratory tests	X		Predose of Period 1. For IV period: 1 hour postdose		X	One hour postdose blood samples will be drawn only for the study period when LY900014 is administered intravenously.
Pregnancy test	X	X			X	For females of childbearing potential only. A serum pregnancy test will be performed at screening and urine pregnancy tests will be performed at all other time points.

Procedures	Screening	Periods 1, 2, 3, 4			FU/ ED[a]	Comments
	Up to Day -28	Day -1	Day 1	Day 2		
Study drug administration			0 hour			Time of study drug administration = 0 hour. Study drug will be administered at approximately the same time on Day 1 of each study period. The site of administration will be determined by the randomization schedule.
Injection-site assessments			0, 1, 4, and 10 hours postdose			Time 0: assessments will occur immediately following the injection.
Insulin lispro PK sampling			2.5, 5, 10, 15, 20, 25, 30, 40, 60, 70, 90, 120, 150, 180, 210, 240, 300, 360, 420, 480, 540, and 600 minutes			Sampling times are relative to the time of study drug administration (time 0).
Treprostinil PK sampling (IV period only)			2.5, 5, 10, and 30 minutes			Only for the study period when LY900014 is administered intravenously. Sampling times are relative to the time of study drug administration (time 0).
C-peptide samples			0, 30, 60, 120, 180, 240, 300, 360, 420, 480, 540, and 600 minutes			Sampling times are relative to the time of study drug administration (time 0).
Blood glucose sampling for euglycemic clamp			Approximately every 10 minutes for approximately 30 minutes before the start of dosing (for baseline measurement). During clamp, sampling occurs every 2.5 minutes for the first 30 minutes; every 5 minutes for 30 to 120 minutes; every 10 minutes for 120 to 480 minutes and every 20 minutes for 480 to 600 minutes.			Sampling times are relative to study drug administration (time 0). Glucose samples should be taken before PK samples. Failure to obtain samples due to clinical issues, such as problems with venous access, will not be considered a protocol violation. Repeat samples for counter-checking of apparent spurious results may be taken where indicated.
Pharmacogenetic sample		X				Period 1 only.
Immunogenicity sample			Predose of Periods 1 and 3		X	Additional samples may be collected if the investigator considers there is a possibility that an adverse event is immunologically indicated.
Discharge from CRU				X		Subjects will be discharged on Day 2.

Abbreviations: CRU = clinical research unit; ECG = electrocardiogram; ED = early discontinuation; FU = follow-up; IV = intravenous; PK = pharmacokinetics.

[a] Procedures should be performed at least 14 days after the last dose of study drug.

Note: The site should schedule activities as appropriate. In cases when several study procedures are scheduled at the same time, the order of priority will be as follows: glucose samples, PK samples, injection-site assessments, vital signs, ECG, clinical laboratory samples. If the investigator decides based on clinical judgment not to dose a subject on a given day (eg, because of low blood glucose), the subject's visit may be rescheduled; any procedures performed in that period may be repeated.

3. Introduction

3.1. Study Rationale

In clinical practice, it is recommended that patients rotate insulin injections and may inject insulin at different tissue sites. The actual site of subcutaneous (SC) injection may influence the absorption pharmacokinetics (PK), as well as, the glucodynamic (GD) effect. The primary objective of this study is therefore to compare the bioavailability of insulin lispro following administration of the commercial formulation of LY900014 at 3 different injection sites SC in the abdominal wall, thigh, and deltoid. In addition, the intravenous (IV) administration will serve as the reference to determine the absolute bioavailability of SC injections into the deltoid, thigh, or abdomen and GD activity.

3.2. Background

The insulin analog, insulin lispro (Humalog®), has been shown to be absorbed more quickly than regular human insulin (Humalog package insert, 2015). In healthy volunteers given SC doses of insulin lispro ranging from 0.1 to 0.4 U/kg, peak serum levels were seen 30 to 90 minutes after dosing (Humalog package insert, 2015). However, the general consensus is that rapid-acting insulin, administered either through pumps or syringes/pen injectors, is still not rapid enough to match carbohydrate absorption profiles, which limits efficacy and dosing flexibility. An ultra-rapid-acting prandial insulin would shift the PK and GD profiles so that they have an even faster onset to better match carbohydrate absorption.

LY900014 is an ultra-rapid-acting insulin lispro formulation with treprostinil (active ingredient in Remodulin®) and other ingredients. LY900014 has an increased early absorption of insulin lispro compared to commercially available insulin lispro (Humalog; Eli Lilly and Company). This formulation involves the novel use of a microdose of treprostinil as an excipient to enhance the absorption of insulin lispro through local vasodilatation rather than as an active pharmaceutical ingredient to elicit a systemic effect.

Treprostinil is a prostacyclin analog, administered either through inhalation (Tyvaso®), as an IV infusion, or as a continuous SC administration for the treatment of symptomatic pulmonary arterial hypertension (PAH) and has been approved in the US since 2002 (Remodulin package insert, 2014). Each of the other excipients (such as sodium citrate and magnesium chloride) in the LY900014 formulation is listed in the US Food and Drug Administration (FDA)'s Generally Recognized as Safe Food Additives database and in the FDA's Inactive Ingredients in Approved Drugs database. Furthermore, the excipient concentration in LY900014 is within the limits identified for approved drug products in the FDA Inactive Ingredients in Approved Drugs database.

Safety and tolerability of LY900014 have been demonstrated in healthy subjects in 3 previous clinical studies, in approximately 71 healthy subjects across a dose range of 7.5 to 30 U. All 4 studies were Phase 1, randomized, subject-blind studies in which the PK and GD of LY900014 and Humalog were evaluated during a euglycemic glucose clamp following SC administration. The total insulin lispro exposure and GD effect were similar for LY900014 and Humalog;

however, LY900014 demonstrated a faster and earlier insulin lispro absorption compared to Humalog.

In addition, data from two Phase 1b studies showed LY900014 was well tolerated in patients with type 1 diabetes mellitus (T1DM; 30 patients) and type 2 diabetes mellitus (T2DM; 30 patients) using multiple daily injections (MDIs). There were no serious adverse events (SAEs) related to study treatment or discontinuations from the studies because of a drug-related adverse event (AE). Small numbers of treatment-emergent adverse events (TEAEs) were reported, and there were no notable increases in these events in relation to any of the LY900014 formulations compared to those in relation to Humalog.

More information can be found in the Investigator's Brochure (IB) of LY900014.

3.3. Benefit–Risk Assessment

Study I8B-MC-ITRT (ITRT) will not offer any direct benefits to the subjects participating in the study. The data from previous studies in healthy subjects and data from studies in patients with T1DM and T2DM have shown that LY900014 was well tolerated and the adverse drug reactions are in keeping with those reported for Humalog.

Potential risks associated with LY900014, derived from the known risks of insulin lispro (Humalog), are hypoglycemia, hypersensitivity reactions (localized allergy and/or systemic allergy), undesirable effects at the injection site (injection-site reactions and lipodystrophy), and peripheral edema (Humalog package insert, 2015).

Notably, across all doses in the Lilly clinical studies that have evaluated treprostinil as a local vasodilator with or without insulin lispro, there was no clinically significant increase in those AEs associated with systemic absorption of treprostinil, as described in the Remodulin package insert (2014). The exposures of treprostinil in LY900014 in this study are expected to be undetectable compared to those observed in the dose ranges previously explored with SC bolus administration of treprostinil and are expected to be substantially lower than those observed in the treatment of PAH.

In preclinical safety pharmacology and toxicity studies, or clinical pharmacology studies involving LY900014 or treprostinil alone, other than known risks associated with Humalog and Remodulin, no additional risks were identified. No known potential risks are associated with the use of small amounts of treprostinil (Remodulin) in the LY900014 formulation.

Additionally, local and systemic toxicity profiles of Humalog and Remodulin do not suggest the potential for additive or synergistic toxicity.

Following administration of the study insulin, subjects will receive IV glucose infusion at a variable rate to maintain euglycemia up to 10 hours after insulin lispro administration. The aim of the clamp procedure is to maintain blood glucose within the normal glycemic range. In addition, the clamp is performed while the subject is inpatient and under the investigator's supervision. These considerations should minimize the risk of hypoglycemia in subjects participating in Study ITRT.

More information about the known and expected benefits, risks, SAEs, and reasonably anticipated AEs of LY900014 are to be found in the IB.

4. Objectives and Endpoints

Table ITRT.1 shows the objectives and endpoints of the study.

Table ITRT.1. Objectives and Endpoints

Objectives	Endpoints
Primary To determine the relative bioavailability of insulin lispro from LY900014 after SC injection into the thigh and deltoid compared to the abdomen in healthy subjects.	AUC(0-∞)
To determine absolute bioavailability of insulin lispro following SC administrations of LY900014 into the thigh, deltoid, and abdomen compared to IV administration of LY900014 in healthy subjects.	Percent of the ratio of AUC(0-∞)
Secondary To evaluate the tolerability of LY900014 in healthy subjects.	Adverse events and injection-site reactions
To evaluate the GD during euglycemic clamp procedure of LY900014 after SC injection into the abdomen, thigh, and deltoid and IV in healthy subjects.	G_{tot} and R_{max}
Exploratory Objectives To evaluate C-peptide levels after administration of LY900014.	C-peptide concentrations
To determine plasma concentrations of the excipient, treprostinil, following administration of a single, IV 15-U dose of LY900014 in healthy subjects.	Plasma concentrations–over-time profile

Abbreviations: AUC(0-∞) = area under the concentration versus time curve from time zero to infinity; GD = glucodynamics; G_{tot} = total amount of glucose infused; IV = intravenous; R_{max} = maximum glucose infusion rate; SC = subcutaneous.

5. Study Design

5.1. Overall Design

This is a Phase 1, single-center, open-label, 4-period, randomized, crossover, up to 10-hour euglycemic clamp study in approximately 28 healthy subjects to compare the insulin lispro PK and GD of LY900014 after SC administration of a 15-U dose at 3 different injection sites and a single IV bolus injection of 15 U. The SC injection sites are the abdomen, thigh, and deltoid.

Subjects will be required to attend the clinical research unit (CRU) on at least 6 occasions:

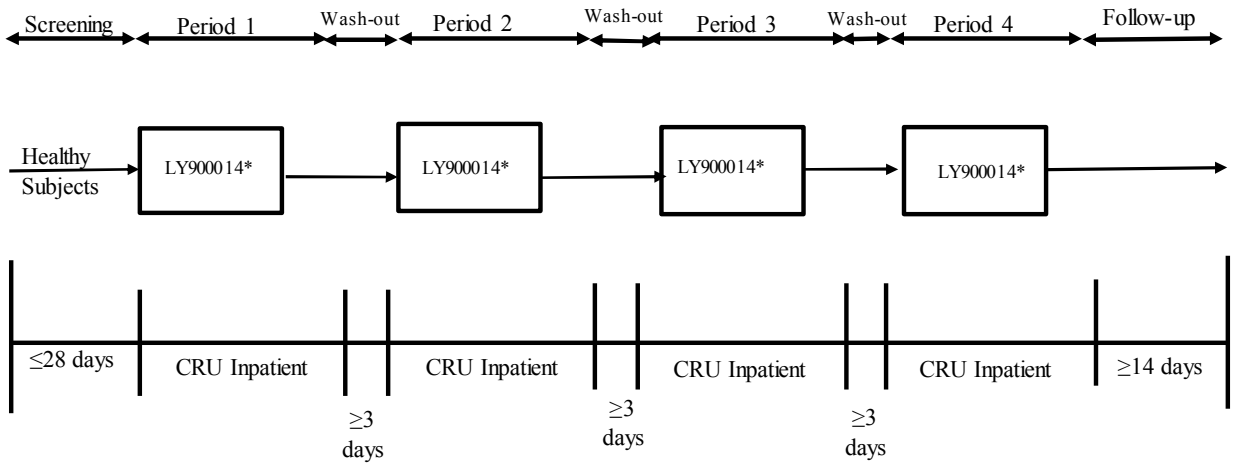
- screening visit (may occur up to 28 days before randomization)
- four treatment visits for the clamp procedure (study Periods 1 to 4) with a wash-out period of ≥3 days between discharge and the next admission to the CRU
- follow-up visit (at least 14 days after last dose), or early discontinuation.

Each subject will be randomly assigned to a treatment sequence and will be administered single doses of 15 U of LY900014 (on 4 occasions).

Subjects will be admitted to the CRU on the evening before each dosing day and will remain in the CRU for the duration of the clamp period and until discharge by the investigator. Subjects are expected to fast for at least 8 hours before each dose. Following dose administration, each subject will undergo an euglycemic clamp procedure of up to 10 hours. Upon completion of the clamp procedures, the subjects will be provided a meal and observed overnight. Subjects will be discharged from the CRU the next day after medical assessments.

Study governance considerations are described in detail in Appendix 3.

Figure ITRT.1 illustrates the study design and Table ITRT.2 presents the treatment sequences.



Abbreviations: CRU = clinical research unit; IV = intravenous; SC = subcutaneous.
*Single dose of LY900014 administered either SC in the abdomen, thigh, deltoid or IV. Up to 10-hour euglycemic clamp procedure after each dose.

Figure ITRT.1. Illustration of study design of Protocol I8B-MC-ITRT.

Table ITRT.2. LY900014 Treatment Sequences

Treatment Sequence	Period 1	Period 2	Period 3	Period 4
1	LY900014 Abdomen	LY900014 Thigh	LY900014 Deltoid	LY900014 IV
2	LY900014 Deltoid	LY900014 Abdomen	LY900014 IV	LY900014 Thigh
3	LY900014 IV	LY900014 Deltoid	LY900014 Thigh	LY900014 Abdomen
4	LY900014 Thigh	LY900014 IV	LY900014 Abdomen	LY900014 Deltoid

Abbreviation: IV = intravenous.
Note: Subcutaneous injection sites are the abdomen, deltoid and thigh.

5.2. Number of Participants

A total of 28 subjects may be enrolled so that at least 22 subjects complete the study. For purposes of this study, a subject completes the study when all scheduled procedures shown in the Schedule of Activities (Section 2) have been completed.

If subjects discontinue from the study before completion of all 4 study periods, replacement subjects may be enrolled up to 28 subjects following agreement between the investigator and the sponsor.

A replacement subject will be assigned the treatment sequence of the discontinued subject and complete that treatment sequence in its entirety.

5.3. End of Study Definition

End of the study is the date of the last visit or last scheduled procedure shown in the Schedule of Activities (Section 2) for the last subject.

5.4. Scientific Rationale for Study Design

A population of healthy subjects is selected based on the likelihood of less physiologic variability in the absence of disease states that may affect multiple organ systems and absence of other confounding factors such as concomitant medications.

The use of a crossover design allows each subject to serve as his or her own control, thereby reducing variability.

A euglycemic clamp technique, the gold standard methodology for assessing insulin action, will be used in this study to provide data on the GD activity of LY900014 (see Section 9.6 for a detailed description of the clamp methodology). Based on previous studies, the 10-hour duration of the euglycemic glucose clamp will allow for a complete assessment of GD activity of LY900014.

Based on the PK properties of treprostinil (half-life associated with the terminal rate constant in noncompartmental analysis [$t_{1/2}$] = approximately 1 hour) and Humalog ($t_{1/2}$ = 0.79 hours), the duration of the washout period between clamp visits (a minimum of 3 days) and the duration between the last dose of study drug and the follow-up visit (at least 14 days) are considered appropriate.

5.5. Justification for Dose

Based on previous studies of both insulin lispro (Humalog) and LY900014, the 15-U dose is within the clinical dose range and should provide measurable PK and GD profiles for insulin lispro. The insulin lispro concentrations in the body resulting from administration of this dose to healthy subjects are anticipated to be measurable over the sampling period. Additionally, it is anticipated that this dose will provide an adequate glucose infusion rate (GIR) for assessment of GD.

The safety, PK, and pharmacology of LY900014 at similar doses and with the similar formulation composition have been assessed in clinical studies in healthy subjects, in patients with T1DM using MDI or insulin pump treatment and T2DM using MDI. In addition, the components of LY900014 have been tested in 3 clinical studies that included the evaluation of the safety, PK, and pharmacology of SC bolus doses of treprostinil (see LY900014 IB).

All tested doses of treprostinil (CCI), insulin lispro, and LY900014 were well tolerated in healthy subjects, patients with T1DM, and patients with T2DM. There were no SAEs related to study treatment in any of the 8 studies. No subject discontinued from the studies because of drug-related AEs. Trial participants in these studies were monitored for changes in vital signs; there were no significant systemic hemodynamic effects of treprostinil at the doses administered based on blood pressure and heart rate. Visual analog scale pain scores showed that the SC injections of insulin lispro plus treprostinil co-formulations and treprostinil alone were well tolerated.

In summary, there were small numbers of TEAEs and injection-site AEs, but there was no clinically significant increase in these or other events compared to placebo or to Humalog and no

clinically significant increase in frequency with higher doses of treprostinil. Notably, at the higher doses of treprostinil, there was no clinically significant increase in those AEs associated with systemic absorption as described in the Remodulin package insert (2014) (that is, headache, diarrhea, nausea, jaw pain, vasodilatation, rash, edema, and hypotension). This is likely attributable to the relatively lower exposures of treprostinil demonstrated in these clinical studies.

It is expected that the treprostinil concentration in the plasma following the 15-U IV bolus administration of LY900014 containing CCI of treprostinil to be below the lower limit of quantitation (LLOQ). As published, the SC infusion of Remodulin has a 100% absolute bioavailability (Remodulin package insert, 2014); thus, it is expected that plasma exposures of treprostinil between IV and SC administrations of LY900014 will be similar. The PK of treprostinil in LY900014 following SC administration of 15 U of LY900014 was assessed in previous studies. In these studies, treprostinil exposure was not detectable for the 15-U dose of LY900014.

6. Study Population

Eligibility of subjects for the study will be based on the results of screening medical history, physical examination, vital signs, clinical laboratory tests, and electrocardiogram (ECG).

The nature of any conditions present at the time of the physical examination and any preexisting conditions will be documented.

Screening may occur up to 28 days prior to enrollment. Subjects who are not enrolled within 28 days of screening may be subjected to an additional medical assessment and/or clinical measurements to confirm their eligibility.

Prospective approval of protocol deviations to recruitment and enrollment criteria, also known as protocol waivers or exemptions, is not permitted.

6.1. Inclusion Criteria

Subjects are eligible for inclusion in the study only if they meet all of the following criteria at screening:

[1] are overtly healthy males or females, as determined by medical history and physical examination

[1a] female subjects:

For female subjects of childbearing potential (defined as not surgically sterilized and between menarche and 1-year postmenopause):

- test negative for pregnancy at the time of screening
- are not lactating
- intend not to become pregnant during the study
- are sexually inactive or have practiced a reliable method of birth control (eg, use of oral contraceptives or levonorgestrel, diaphragms with contraceptive jelly, cervical caps with contraceptive jelly, condoms with contraceptive foam, intrauterine devices, partner with vasectomy, or abstinence) for at least 6 weeks prior to screening
- agree to continue to use a reliable method of birth control (as determined by the investigator) until the end of the study

Female subjects not of childbearing potential due to surgical sterilization (at least 6 weeks after surgical bilateral oophorectomy with or without hysterectomy or at least 6 weeks after tubal ligation) confirmed by medical history or due to menopause.

- Menopausal women include women with either

1) spontaneous amenorrhea for at least 12 months, not induced by a medical condition such as anorexia nervosa and not taking medications during the amenorrhea that induced the amenorrhea (eg, oral contraceptives, hormones,

gonadotropin-releasing hormone, antiestrogens, selective estrogen receptor modulators, or chemotherapy)

or

2) spontaneous amenorrhea for 6 to 12 months and a follicle-stimulating hormone level consistent with menopausal state

[2] are between 21 and 65 years of age at the time of screening, inclusive.

[3] have a body weight of ≥45 kg, and body mass index (BMI) of 18.0 to 30.0 kg/m^2, inclusive

[4] have clinical laboratory test results within normal reference range for the population or CRU, or results with acceptable deviations that are judged to be not clinically significant by the investigator.

[5] have venous access sufficient to allow for blood sampling, IV administration, and clamp procedure as per the protocol.

[6] are nonsmokers, have not smoked for at least 6 months before entering the study, and agree not to smoke (cigars, cigarettes, or pipes) or use smokeless tobacco or nicotine products for the duration of the study.

[7] are reliable and willing to make themselves available for the duration of the study and are willing to follow study procedures.

[8] are able and willing to give signed informed consent.

6.2. Exclusion Criteria

Subjects will be excluded from study enrollment if they meet any of the following criteria at screening and/or enrollment:

[9] are CRU personnel directly affiliated with this study and their immediate families. Immediate family is defined as a spouse, biological or legal guardian, child, or sibling.

[10] are Lilly employees.

[11] are currently enrolled in a clinical study involving an investigational product (IP) or any other type of medical research judged not to be scientifically or medically compatible with this study.

[12] have participated, within the last 30 days, in a clinical study involving an IP. If the previous IP has a long half-life, 5 half-lives, or 3 months (whichever is longer) should have passed.

[13] have previously completed or withdrawn from this study.

[14] have known allergies to treprostinil or insulin lispro, related compounds, or any components of the formulation, or history of significant atopy.

[15] have an abnormality in the 12-lead ECG that, in the opinion of the investigator, increases the risks associated with participating in the study.

[16] have an abnormal blood pressure and/or pulse rate as deemed to be clinically significant by the investigator.

[17] have a history or presence of cardiovascular, respiratory, hepatic, renal, gastrointestinal, endocrine, hematological, or neurological disorders capable of significantly altering the absorption, metabolism, or elimination of drugs; of constituting a risk when taking the IP; or of interfering with the interpretation of data.

[18] have a history or presence of psychiatric disorders as deemed clinically significant by the investigator.

[19] regularly use known drugs of abuse.

[20] show evidence of human immunodeficiency virus (HIV) infection, and/or positive human HIV antibodies.

[21] show evidence of hepatitis B and/or positive hepatitis B surface antigen.

[22] show evidence of hepatitis C and/or positive hepatitis C antibody.

[23] use over-the-counter or prescription medication within 7 to 14 days, respectively, prior to dosing (apart from vitamin/mineral supplements, occasional paracetamol, thyroid replacement medication, or birth control methods) and throughout the study period. If this situation arises, inclusion of an otherwise suitable subject may be at the discretion of the investigator and sponsor.

[24] have donated blood or have blood loss of more than 450 mL within the previous 3 months of study screening.

[25] have an average weekly alcohol intake that exceeds 21 units per week (males) and 14 units per week (females), or are unwilling to stop alcohol consumption from 24 hours prior to dosing and until discharge from the CRU (1 unit = 12 oz or 360 mL of beer; 5 oz or 150 mL of wine; 1.5 oz or 45 mL of distilled spirits).

[26] in the opinion of the investigator or sponsor, are unsuitable for inclusion in the study.

Additional inclusion criterion:

[27] have a hemoglobin level of ≥12.5 g/dL at screening.

6.3. Lifestyle and/or Dietary Requirements

Throughout the study, subjects may undergo medical assessments and review of compliance with requirements before continuing in the study.

6.3.1. Meals and Dietary Restrictions

For all treatment periods subjects will fast for at least 8 hours prior to dosing and until the glucose clamp procedure is completed, after which subjects will receive a meal. Water can be consumed freely during this time.

When not a resident at the CRU, subjects will be encouraged to follow their normal diets.

6.3.2. Caffeine, Alcohol, and Tobacco

No alcohol will be allowed at least 24 hours before each dose and for the duration of each CRU visit.

Subjects should refrain from caffeine-containing food/beverages (eg, cola, chocolate, Milo, tea, and coffee) for at least 12 hours before each dose and throughout the duration of each CRU visit.

Smoking (cigars, cigarettes, or pipes) and the use of smokeless tobacco will not be permitted during the study.

6.3.3. Activity

Subjects are encouraged to maintain their regular exercise habits; however, they should not undertake vigorous or prolonged exercise within 48 hours prior to dosing. These subjects will be excluded from this study, as judged by the investigator to prevent interference with study results. After dosing, subjects should remain recumbent or sitting in the CRU until the end of the glucose clamp.

Movement will be restricted to retain the integrity of connections to the infusion(s) and the study procedures.

6.4. Screen Failures

Individuals who do not meet the criteria for participation in this study (screen failure) may not be re-screened.